Santosh Sai Gowtham Pasala

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PROFESSIONAL SUMMARY

AI Software Engineer with 4+ years of experience building scalable web applications and AI-powered systems. Specializing in Python, Node.js, React.js, and AWS with hands-on expertise in LLM integration, RAG pipelines, and production-grade monitoring. Shipped AI and full-stack features under high concurrency.

EXPERIENCE

Software Engineer | Geico | San Jose, CA — Jul 2024 – Present

- Architected a multi-agent email processing system using **LangGraph** orchestration with intent classification, routing logic, and tool execution handling 1K+ customer service emails daily with 92% accuracy.
- Implemented production-grade **RAG pipeline** with vector embeddings and chunking strategies to retrieve relevant knowledge base articles, reducing average response time from 4 hours to 12 minutes.
- Optimized secure PII/PCI masking layer using Microsoft Presidio + custom regex patterns to anonymize sensitive customer data before **LLM processing**, ensuring SOC 2 compliance.
- Developed **React.js** admin dashboard for monitoring agent performance, reviewing flagged responses, and implementing human-in-the-loop approval workflows.
- Developed **FastAPI** microservices for email parsing (.eml), intent classification, and response generation with async request handling, deployed via **Docker/Kubernetes** for 99.9% uptime.
- Integrated **LangSmith** observability to track agent reliability, tool call success rates, and response accuracy across multi-turn conversations, identifying and fixing 15+ edge cases.
- Designed **PostgreSQL** schema for conversation state management, enabling context-aware responses across email threads and A/B testing of prompt templates.
- Leveraged **GitHub Copilot** and **Cursor** to accelerate development of custom LangChain tools and LangGraph state machines, reducing boilerplate code by 40% while maintaining test coverage above 85%.

Software Analyst | Deloitte | Hyderabad, India — Aug 2021 – Aug 2022

- Delivered scalable full-stack solutions using **Python**, **FastAPI**, **SQLAlchemy**, and **React**, building high-performance enterprise applications.
- Designed and developed **FastAPI**-based microservices following scalable architecture principles and design patterns such as Singleton, Factory, and DAO.
- Built backend services to process and transform **JSON** data for microservices communication, ensuring efficient data exchange and integration.
- Developed robust applications using Core **Python**, multithreading, asynchronous programming (async/await), and RESTful web services.
- Leveraged **SQLAlchemy** ORM for efficient database interactions and optimized query performance.
- Implemented RESTful APIs using **FastAPI**, enabling seamless data ingestion, processing, and storage of JSON payloads.
- Designed dynamic, responsive, and cross-browser-compatible user interfaces using **JavaScript**, **jQuery**, and **ReactJS**.
- Wrote unit and integration tests using **PyTest** and unittest, improving code quality and reducing production defects.
- Provided production support by debugging, monitoring, and resolving issues, ensuring high system availability and minimal downtime.

Software Engineer | Virtutech | Hyderabad, India — May 2020 – Jul 2021

- Led the development and launch of a responsive e-commerce platform using **React**, **Redux**, **TypeScript**, and **jQuery**, increasing customer engagement by 35% while ensuring cross-browser compatibility.
- Designed and implemented scalable backend services using **Node.js** and **Python** (FastAPI), developing RESTful APIs for real-time data exchange, improving data retrieval efficiency by 50%, and overall system performance.
- Developed and executed comprehensive test suites using Jest, Enzyme, and Python (PyTest), achieving 90% code coverage and improving code reliability and debugging efficiency.
- Automated CI/CD pipelines using AWS CodePipeline, CodeBuild, Docker, and Python-based automation scripts, reducing deployment time by 40% and enabling continuous delivery with 99.9% system uptime.

PROJECTS

Multi-Agent Research System: AI-Powered Deep Research Platform | GitHub

- Built a multi-agent research platform using **LangGraph**, orchestrating 5 specialized agents with a conditional feedback loop for iterative refinement. Integrated semantic search and **Pinecone** vector memory with similarity-based caching.
- Implemented a **FastAPI** backend with SSE streaming for real-time agent progress, multi-layered safety guards (**LlamaGuard** prompt injection detection, PII filtering). Monitored with **LangSmith** for end-to-end LLM observability.

OmniQuery: Agentic SQL-RAG for Business Data | GitHub

- Built a full-stack GenAI agentic system using LlamaIndex and Gemini that autonomously routes user queries to either a SQL database for structured business metrics or a vector store for semantic search over unstructured documents.

TECHNICAL SKILLS

Core: JavaScript, TypeScript, Python, Java, React.js, Node.js, Express.js, Next.js, Spring Boot, Angular, FastAPI.

Data & AI: PostgreSQL, MongoDB, DynamoDB, Redis, LangChain, LangGraph, PGVector, Pinecone, LangSmith.

Cloud & DevOps: AWS (Lambda, S3, EC2, CloudFront, SQS, SNS), Docker, Kubernetes, Jenkins, GitHub Actions.

Monitoring & Testing: Grafana, Prometheus, Datadog, ELK Stack, k6, Selenium, JUnit, Postman.

Also familiar: C/C++, C#, Go, Vue.js, Django, GraphQL, Kafka, Cassandra, Snowflake, GCP, Azure.

EDUCATION

Master of Science in Software Engineering | San Jose State University — Aug 2022 – May 2024

Bachelor of Technology in Computer Science | VNR VJIET — Aug 2016 - May 2020